

# A Win Probability Strategy for the Power Play in Mixed Doubles Curling

Trey Elder Samen Hossain Jonathan Pipping

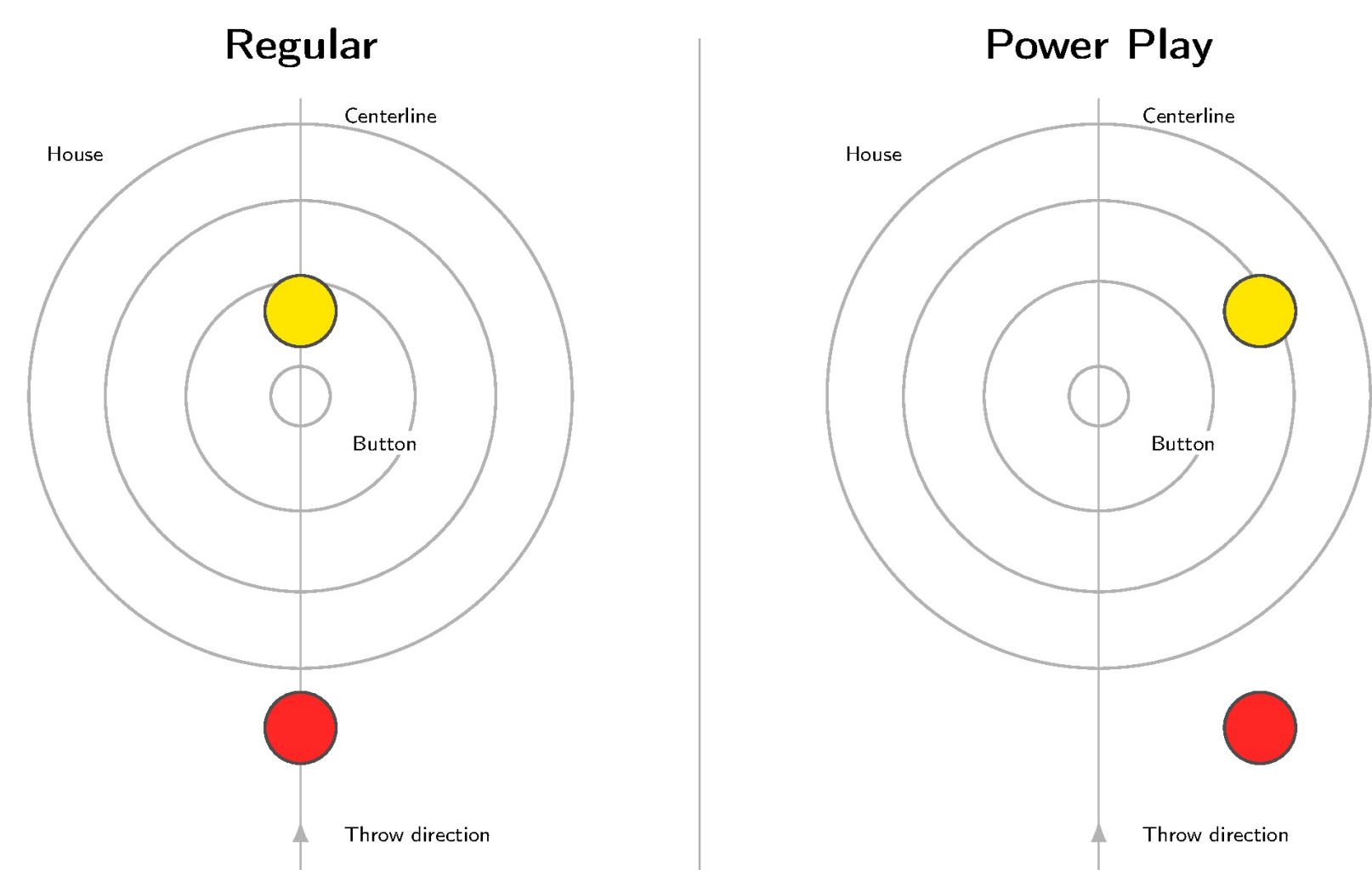
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## The Question

In mixed doubles curling, each team gets **one Power Play per game**, which shifts pre-positioned stones to open the scoring lane. Since teams only have one to use, the big question is:

*When should you deploy it?*

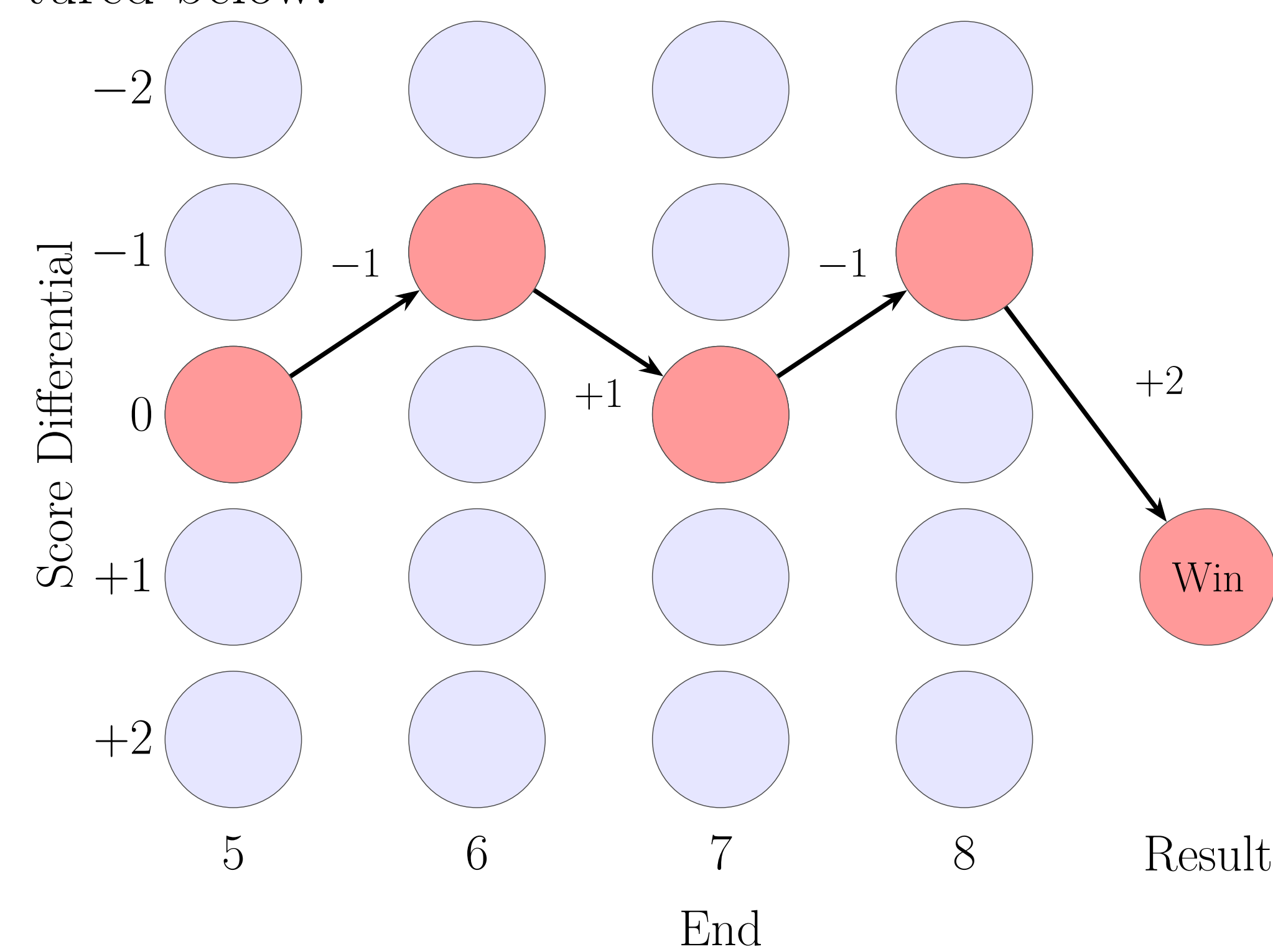
## The Power Play



Regular setup (left) vs. Power Play (right).

## Modeling Approach

Reasoning based on expected points can conclude “use PP earlier” because PP adds points in any end. But a point in end 3 is **not** worth the same as a point in end 8: optimizing EP ignores the **option value** of saving a scarce resource for a higher-leverage moment. Instead, we model the match as a **finite-horizon MDP** [1] with states  $s_e = (e, d_e)$ , as pictured below.



## Mathematical Solution

- Transitions:** A multi-class XGBoost model conditions on hammer, PP availability, and relative team strength (Elo) to estimate the end-outcome distribution  $P(\Delta_e | s_e, a)$  from **5,274** end-states, **344** games [2].
- Solve:** Backward induction yields the optimal stopping rule:

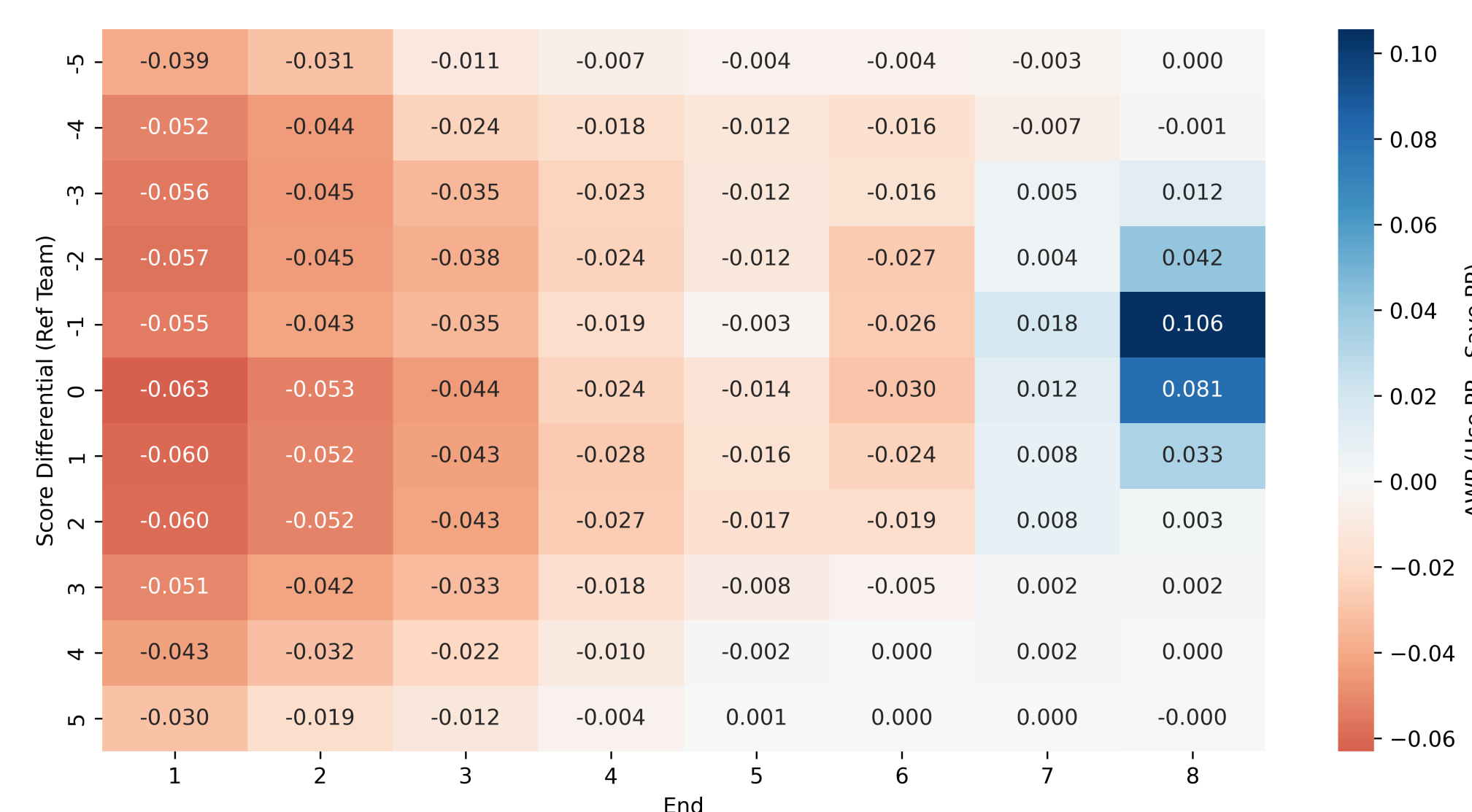
$$\Delta WP = V(s_e | \text{use}) - V(s_e | \text{save}) > 0$$

*i.e.*, use PP if it increases the estimated win probability relative to saving for later.

## Key Result

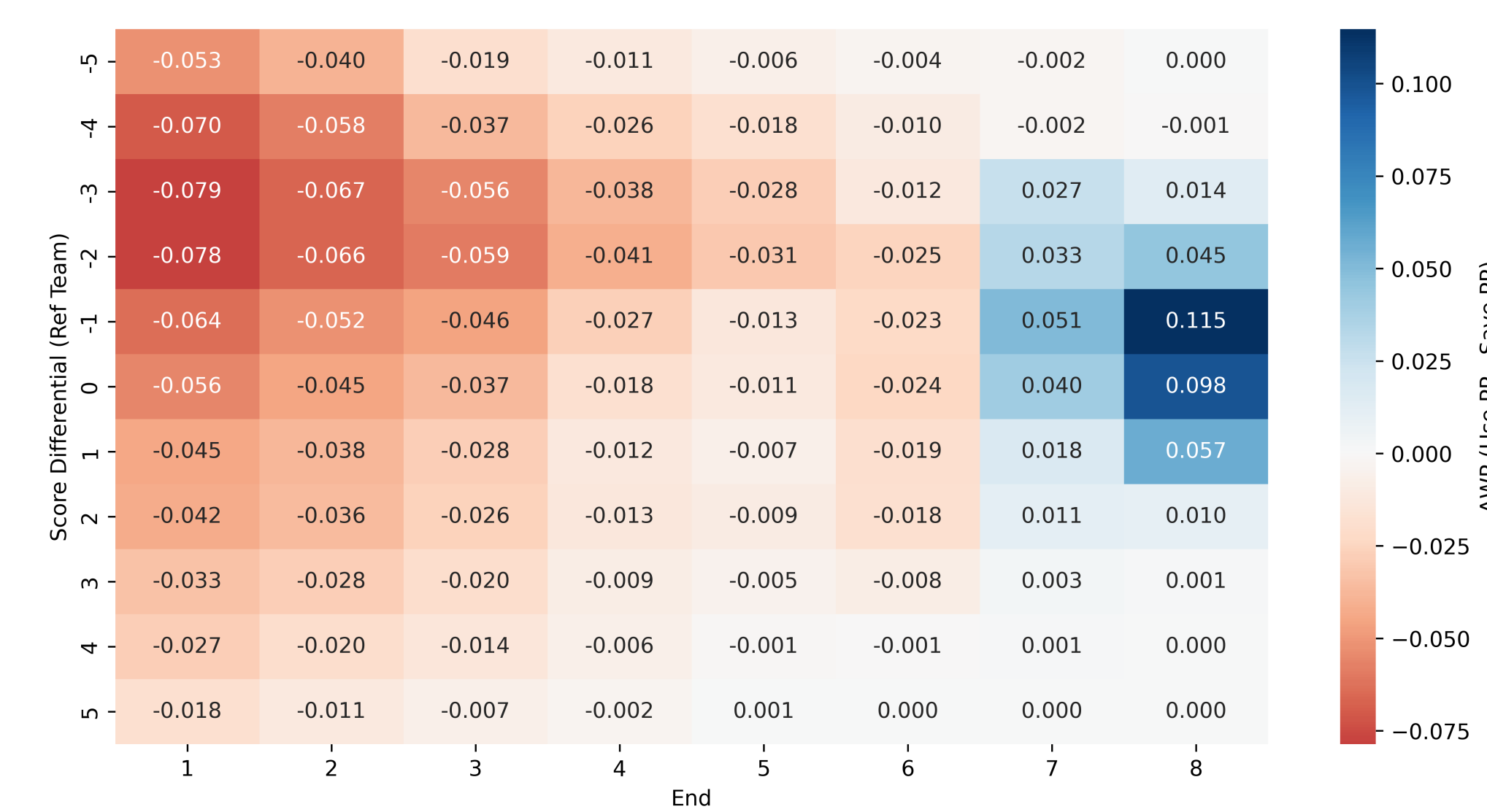
End 8, down 1: PP adds **+10.6%** win probability. Tied: **+8.1%**. Earlier in the game, the **option value of saving** almost always dominates.

## Optimal Policy: Opp PP Avail.



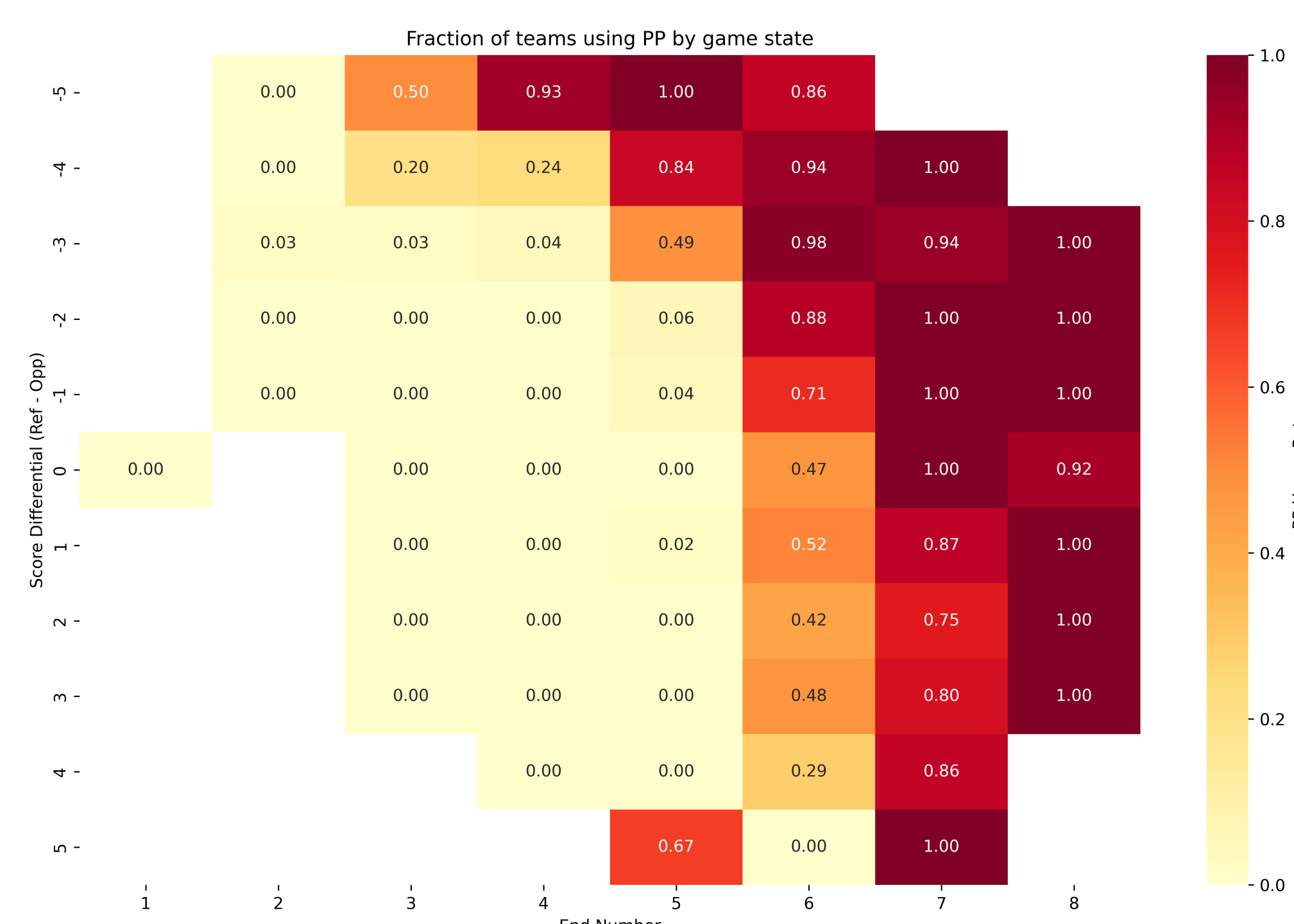
$\Delta WP$  by end and score. Blue = use now; red = save.

## Optimal Policy: Opp PP Used



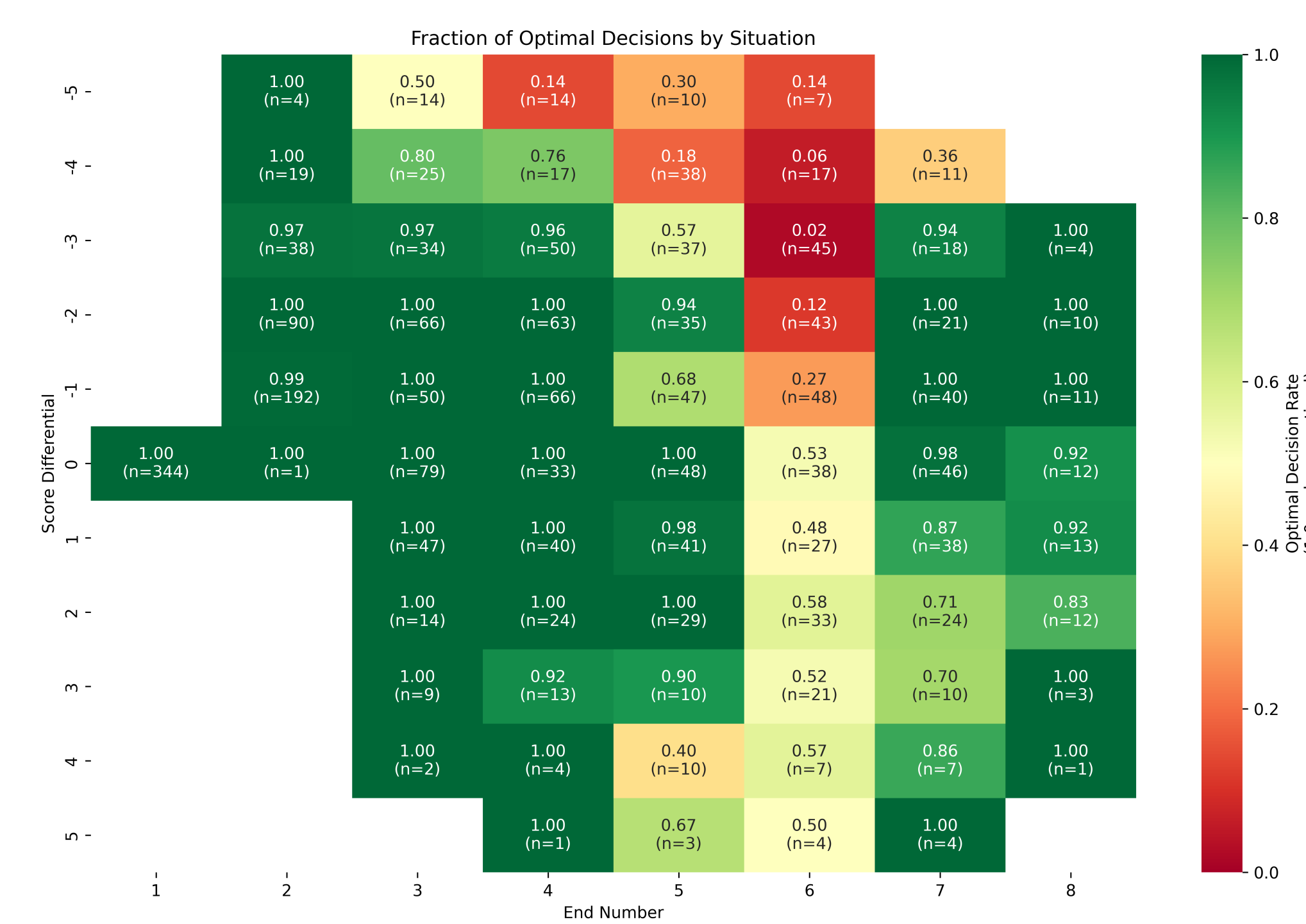
PP is worth *more* when the opponent has already used theirs.

## Observed PP Usage



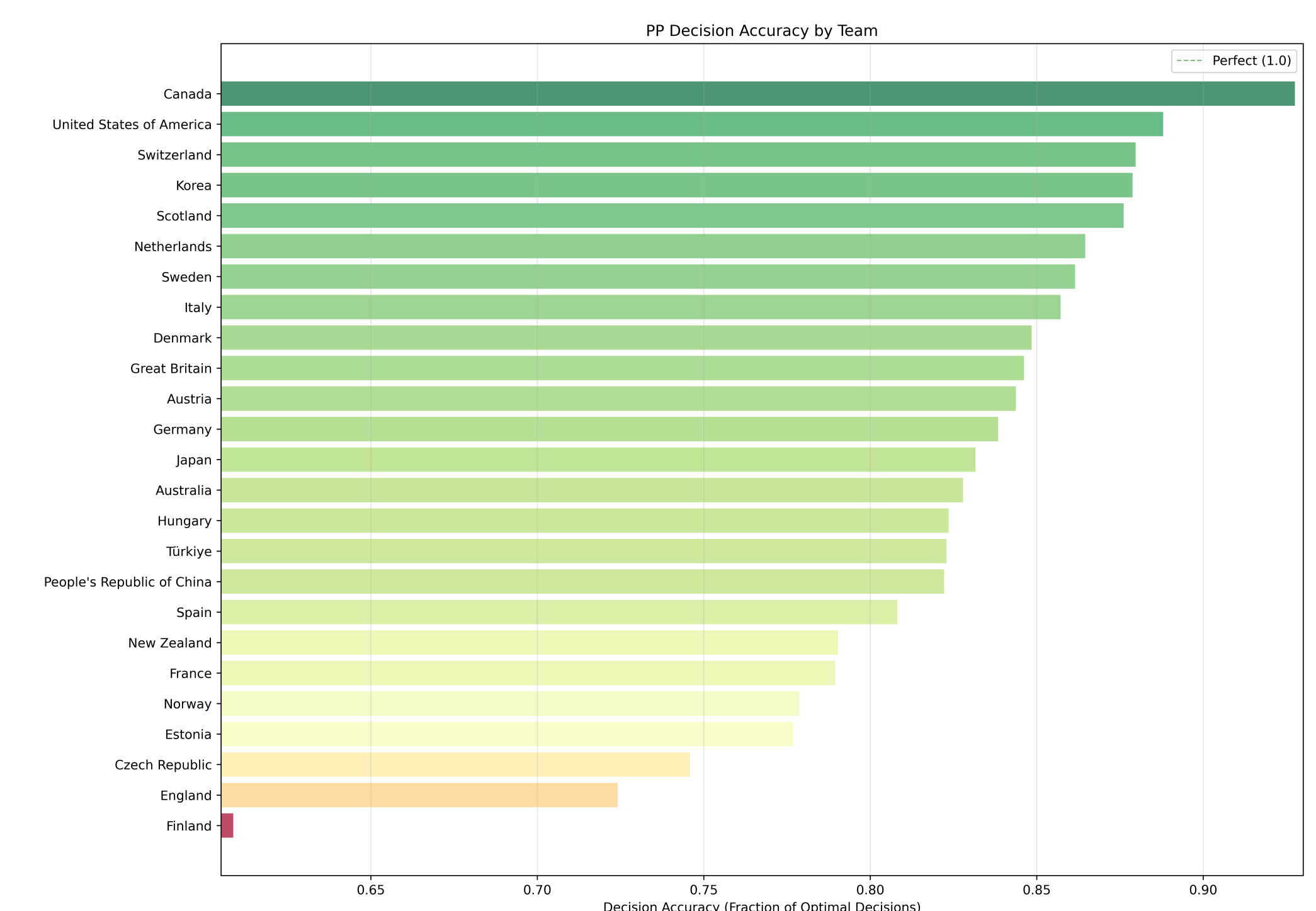
When teams actually deploy PP. Most used in ends 5–8.

## Decision Accuracy vs. Optimal

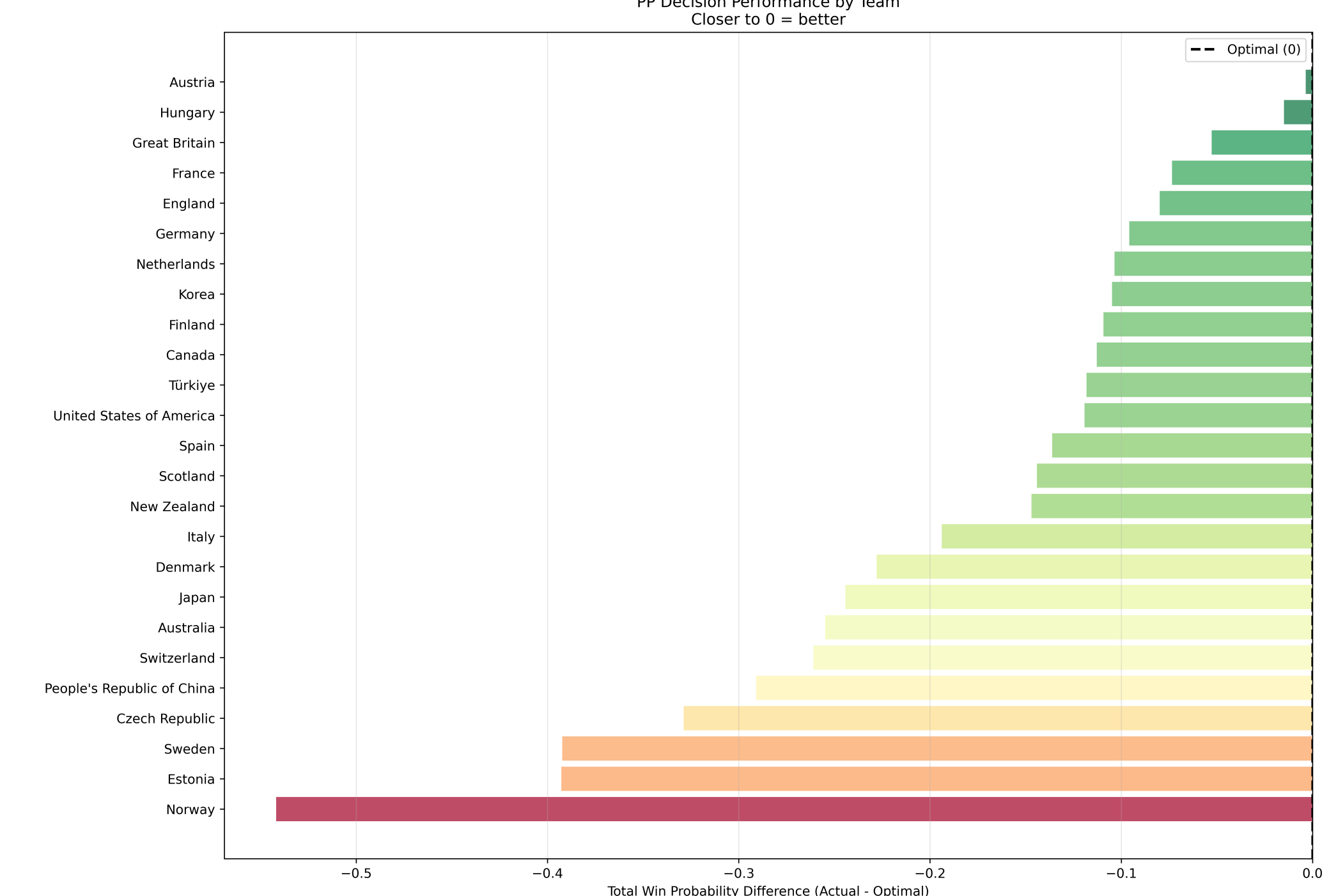


Biggest departures: early use in end 6 (up to 1.8% WP loss).

## Team-Level Performance



Decision accuracy by team ( $\geq 5$  decisions).



Total WP impact of PP timing decisions by team.

## Practical Guidance

**Ends 1–6:** Default to **saving**: option value is large.  
**Ends 7–8:** Use in almost all game states.

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## References

- [1] Kostuk et al., “Modelling curling as a Markov process,” *EJOR*, 2001.  
[2] CSAS 2026 Data Challenge, [stats.org/events/csas2026](https://stats.org/events/csas2026)